Minimum rival covers for 108 fixed RNA-Undesign motif instances

Tatsuhiko Takimoto
Independent researcher

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Abstract

A collection of rival RNA structures can certify that a target is never the unique minimum-energy structure. We determine the smallest size of such a collection for 108 fixed motif instances from RNA-Undesign. With fixed boundaries and ViennaRNA's Turner 2004 model at 37 °C, the minima are between one and six, with a sum of 310 over the instances. Each optimum has an upper cover and a finite lower witness set. The accompanying checker reconstructs the candidate structures from a stated grammar, checks lower bounds, and exhaustively verifies upper coverage; the certificates, the checker and the stored output of one complete check run are identified by SHA-256 in Section 4. These are finite, model-specific results; they do not assert a bound for arbitrary RNA.

1 Instances, boundaries and energy model

The source is the pinned RNA-Undesign CSV, at `data/motifs_maxlen14_no_external/results.uniq.csv` in Git commit `8a8de22`. Of its 1,805 data rows, the 108 rows with `N_Rival` ≥ 2 are selected. The row identifiers in Tables 1–2 are one-based data-row numbers, excluding the header. RNA-Undesign supplies the motifs and the rival-based framework [1, 2]; the task here is to minimize the number of rivals over an explicitly specified candidate set.

A motif has an outer base pair and zero or more boundary pairs denoted by `(*)`. Each boundary is an indivisible closed child. For energy evaluation, it is replaced by a hairpin with structure `(...)` and three fixed internal A letters. The outer and boundary pairs are preserved in every candidate. The internal A letters are never available as new pairing endpoints.

Let $\mathcal{X}(m)$ contain all sequences compatible with the target m: every target pair, including outer and boundary pairs, allows AU, UA, GC, CG, GU or UG; every other unfixed unpaired position allows A, C, G or U. Thus, with p target pairs and u unfixed unpaired positions,

$$|\mathcal{X}(m)| = 6^p 4^u.$$

Candidate set. Write the motif in compressed form: a string $t = t_0 t_1 \cdots t_{n-1}$ over the symbols `(`, `)`, `.` and $*$, where each boundary `(*)` becomes the single atomic symbol $*$. The positions 0 and $n-1$ form the outer pair. A candidate is a string s of the same length such that

1. $s_0 =$ `(` and $s_{n-1} =$ `)`, and these two positions pair with each other (the outer pair is preserved);
2. $s_i = *$ exactly when $t_i = *$ (every boundary is preserved and is never split; its three internal A letters are not positions of s, so they never pair);
3. the remaining symbols of s are `(`, `)` or `.` and form a balanced, hence noncrossing, bracket structure;
4. every pair of s, including the outer pair, encloses at least one directly nested child or at least three directly enclosed `.` symbols. A direct child is a pair of s or a $*$ that is not inside another pair of s within this loop.

Then $\mathcal{R}(m)$ is the set of such strings other than t. New pairs may enclose one or more boundaries, so the containment of boundaries inside other pairs may change; only the outer pair and the boundary pairs themselves are fixed. Rule 4 counts hairpin length in compressed coordinates: a $*$ is a child, not three unpaired letters. For example, the synthetic motif `(.(*)...)` has compressed form `(.*...)` and exactly three candidates, `((*)..)`, `((*.).)` and `((*..))`; the string `(.*(.))` is excluded by Rule 4. For energy evaluation every $*$ is expanded

back to `(...)` with internal letters AAA. The checker enumerates $\mathcal{R}(m)$ with a stack automaton implementing exactly Rules 1–4; its candidate counts are those in Tables 1–2.

Energy $E(x, r)$ is evaluated using ViennaRNA 2.7.2 [4, 5], with the Turner 2004 parameter set that ships with that version, loaded by `RNA.params_load_RNA_Turner2004()`. The model details are set explicitly to temperature 37 °C, `dangles=2`, `special_hp=1`, `noLP=0`, `noGU=0`, `noGUclosure=0`, `min_loop_size=3`, `circ=0`, `gquad=0`, `logML=0`, salt 1.021 M and `energy_set=0`; all other settings are the ViennaRNA 2.7.2 defaults. The checker refuses to run if the installed version or any of these values differs. The integer `eval_structure_pt` API reports units of 0.01 kcal/mol. No floating-point tolerance is used in the cover comparison.

2 The finite minimum-cover claim

A rival $r \in \mathcal{R}(m)$ covers $x \in \mathcal{X}(m)$ if all its pairs are compatible with x and

$$E(x, r) \leq E(x, m).$$

Ties count: this prevents m from being the *unique* minimum-free-energy structure. The quantity determined is

$$k^*(m) = \min\{|C| : C \subseteq \mathcal{R}(m),\ \forall x \in \mathcal{X}(m)\ \exists r \in C :\ r \text{ covers } x\}.$$

An upper certificate is a list U_m of k distinct candidates covering all of $\mathcal{X}(m)$. A lower certificate is a nonempty finite set $W_m \subseteq \mathcal{X}(m)$ such that no set of at most $k-1$ candidates covers W_m.

Proposition 1. *If these upper and lower certificates hold, then $k^*(m) = k$.*

Proof. The upper certificate gives $k^*(m) \leq k$. A cover of $\mathcal{X}(m)$ also covers its subset W_m, so the lower certificate gives $k^*(m) \geq k$. For $k = 1$, the nonempty sequence space already excludes an empty cover. □

For a lower check, enumerate all candidates and evaluate their coverage on W_m. Represent each coverage set by a bit mask. Let $S_0 = \{0\}$ and $S_d = S_{d-1} \cup \{a \vee c : a \in S_{d-1},\ c \text{ a candidate mask}\}$, so earlier states, including 0, are kept. Then S_d is exactly the set of unions of at most d candidate masks. Absence of the full witness mask from S_{k-1} proves the lower bound. Reusing a candidate does not change a union, so the procedure includes every subset of at most $k-1$ candidates. No optimization solver or search implementation is required.

For an upper check, enumerate all $6^p 4^u$ sequences and require at least one compatible rival in U_m with energy no larger than the target. The candidates are regenerated by a stack grammar that treats boundaries atomically; the certificates do not supply a purportedly complete candidate list.

3 Results

The 108 minima and the candidate and sequence counts are listed in Tables 1–2. There are 12,235 candidates in total across the instance-specific candidate sets, and 300,119,040 sequence assignments. The distribution is

Minimum k^*	1	2	3	4	5	6
Instances	10	31	40	19	6	2

The sum of the instance-specific minima is 310. It is not a claim of 310 distinct rivals forming one cover for all motifs. The lower witness sets contain one to six sequences; their size and the certificate byte size are not minimized. The original CSV's rival counts are outputs of its search procedure, not minimum-cardinality claims being corrected here.

4 Reproduction and scope

The certificates and checker are in version 1.0.0 of the package at `https://github.com/tkmt-lab/rna-undesign-minimum-rivals` (Git tag `v1.0.0`); the archived copy of that release and its DOI are linked from the repository page. The files on which the results depend, and the two files recording the complete check run described below, are identified by SHA-256:

data/certificates.json	c0e7ced0ea783467fef0927be667cb62a819a3a83a03fa35378ddd700e4c26b1
data/results.csv	74bce882a58445fef972dc603b5a0245a3a5d29729a6a63a959234b2ebfffcc4
checkers/geometry.py	94d83598e0842c74a9d79b3581aad5a526cb9e3f5252a995057cc3e5aa623e0d
checkers/verify.py	c01320dbe263a90ec37b0bed003839c993508d1f5ace8415c07d1d231a9414a7
validation/full-check.json	53be272ff0ef1ba38d924b5b8e1fd71608c398e76d72cd36d33e85759ff76977
validation/full-check-meta.json	427e08c3a6f68313d07cb402956e8748828858e9e3727381c989d613a7c444b7

For each instance the certificate file gives the k upper rivals U_m and the lower witnesses W_m. The package also contains the source-fetch tool. The original CSV and its motif strings are not redistributed. A user obtains the pinned CSV separately; the tool accepts it only if its SHA-256 is

`a06d6cc495c36c50b4292e260ebc89d54d3dbd422b99fd2ebfae66431ff59041`.

Each certificate includes the SHA-256 of the exact decoded `Motif` field, encoded as UTF-8 without an added newline or normalization. Stored rivals use the compressed boundary notation; witness sequences use the expanded coordinates with fixed AAA. The checker requires the exact 108 selected row identifiers before reconstructing the motifs.

The default smoke run checks all candidate counts, sequence counts and lower certificates, and exhaustively checks upper covers for m0081, m0335 and m1780, totaling 539,136 sequences. The `--full` option checks every upper cover and all 300,119,040 sequences. The smoke run alone is not a full upper-coverage verification. The output of a complete `--full` run of the packaged checker on these files is stored in `validation/full-check.json`; it lists, for every instance, the lower-bound result and the number of sequences checked for the upper cover, and `validation/full-check-meta.json` records the SHA-256 of each input file of that run. A hash fixes these record files; it does not by itself prove that the run took place, which a reader confirms by rerunning `--full`. The packaged checker derives from the research-side certificate grammar; it is not a new blind independent review. The `VERIFICATION-RECORD.md` file separates an earlier complete certificate check by a separately written checker from the package checks and records evaluator dependencies.

The mathematical claim concerns this fixed sequence space, candidate grammar, boundary treatment and parameter set. It does not cover unconstrained rival structures, other thermodynamic parameters, pseudoknots, measured molecular behavior or all RNA motifs. In particular, we do not claim that the minimum with fixed outer and boundary pairs equals a minimum over unconstrained rival structures, and we have not tested robustness to boundary interiors other than the three fixed A letters. Computational certificate checks establish the stated finite claims relative to their implementation and energy evaluator; the software is not formally verified.

Rival-based certificates of undesignability and the enumeration of minimal undesignable motifs are prior work [1, 2, 3]. Minimality under inclusion of motifs is different from the minimum *number of rivals* for a fixed motif. Our contribution is the fixed collection of optimum values and their upper and lower certificates. No priority claim is made. A possible next question is how these minima change under a separately specified boundary or parameter change; the present certificates do not answer it.

AI systems assisted with the calculations, certificate checks, literature comparison, code and writing. Tatsuhiko Takimoto is responsible for the report. This draft has not undergone human peer review.

References

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[2] T. Zhou, A. Malik, W. Y. Tang, D. H. Mathews and L. Huang. Scalable and Interpretable Identification of Minimal Undesignable RNA Structure Motifs with Rotational Invariance. RECOMB 2025. arXiv:2402.17206.

[3] T. Zhou, A. Malik, W. Y. Tang, D. H. Mathews and L. Huang. Theory, Algorithms, and Applications for Identification of Undesignable RNA Secondary Structures and Motifs. J. Comput. Biol. 33(1):134–167, 2026. doi:10.1177/15578666251398551.

[4] R. Lorenz et al. ViennaRNA Package 2.0. Algorithms Mol. Biol. 6:26, 2011. doi:10.1186/1748-7188-6-26.

[5] ViennaRNA Package, version 2.7.2. Python distribution `ViennaRNA==2.7.2`, https://pypi.org/project/ViennaRNA/2.7.2/.

Instance results (1/2)

ID	k^*	Candidates	Sequences
m0025	3	64	5,308,416
m0039	4	24	124,416
m0044	3	56	1,990,656
m0081	4	34	82,944
m0137	2	61	3,538,944
m0139	2	159	5,308,416
m0144	3	40	497,664
m0155	1	118	5,308,416
m0208	5	273	14,155,776
m0215	2	118	14,155,776
m0232	2	273	14,155,776
m0237	1	273	14,155,776
m0254	2	92	1,327,104
m0265	4	92	1,327,104
m0295	6	110	1,990,656
m0314	2	90	1,990,656
m0317	3	49	497,664
m0335	2	14	124,416
m0353	3	19	124,416
m0401	4	126	497,664
m0420	4	159	5,308,416
m0470	2	65	746,496
m0489	2	96	1,990,656
m0501	2	69	1,990,656
m0525	2	56	1,990,656
m0562	3	56	1,990,656
m0564	2	21	497,664
m0569	3	34	746,496
m0571	4	96	746,496
m0618	3	110	1,990,656
m0624	2	21	221,184
m0630	3	69	746,496
m0641	4	220	5,308,416
m0653	3	90	1,990,656
m0656	3	34	497,664
m0664	3	254	1,990,656
m0682	3	65	1,990,656
m0692	3	34	497,664
m0695	3	19	746,496
m0716	3	90	746,496
m0717	3	110	1,990,656
m0719	4	126	497,664
m0732	4	220	1,990,656
m0756	3	69	884,736
m0779	4	34	82,944
m0812	2	64	5,308,416
m0820	3	90	1,990,656
m0839	3	159	5,308,416
m0855	3	110	746,496
m0857	2	92	1,327,104
m0866	3	254	5,308,416
m0885	1	159	1,990,656
m0916	2	56	746,496
m0920	2	40	497,664

Table 1: First 54 selected source rows.

Instance results (2/2)

ID	k^*	Candidates	Sequences
m0941	1	92	5,308,416
m0968	2	14	746,496
m0973	2	34	1,990,656
m0998	2	159	5,308,416
m1006	3	118	1,990,656
m1016	4	126	1,327,104
m1022	2	65	1,990,656
m1053	3	92	5,308,416
m1056	2	118	5,308,416
m1067	4	220	1,990,656
m1075	2	126	1,327,104
m1082	1	61	1,327,104
m1086	4	110	746,496
m1099	5	126	1,327,104
m1142	3	90	746,496
m1180	4	126	497,664
m1208	3	69	1,990,656
m1213	4	220	1,990,656
m1218	4	90	746,496
m1220	2	90	746,496
m1230	3	24	124,416
m1271	2	159	5,308,416
m1293	3	110	1,990,656
m1385	5	273	5,308,416
m1401	3	118	1,990,656
m1419	3	96	746,496
m1448	3	90	746,496
m1473	3	96	1,990,656
m1491	3	45	3,538,944
m1495	2	118	5,308,416
m1508	2	64	1,990,656
m1522	2	34	497,664
m1528	4	254	1,990,656
m1602	6	132	1,327,104
m1616	3	90	1,990,656
m1619	5	273	14,155,776
m1626	2	65	1,990,656
m1629	3	65	1,990,656
m1647	1	159	1,990,656
m1662	5	273	14,155,776
m1663	3	254	1,990,656
m1664	3	254	1,990,656
m1668	3	92	5,308,416
m1680	3	132	3,538,944
m1702	2	159	5,308,416
m1704	1	92	1,327,104
m1718	4	126	497,664
m1740	4	273	5,308,416
m1760	1	159	1,990,656
m1780	1	34	331,776
m1791	5	132	3,538,944
m1797	1	132	3,538,944
m1802	3	254	1,990,656
m1803	2	132	3,538,944

Table 2: Remaining 54 selected source rows.